A Two-Grid Method for Mixed Finite-Element Solution of Reaction-Diffusion Equations

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We present a scheme for solving two-dimensional, nonlinear reaction-diffusion equations,

$$s\frac{\partial p}{\partial t} - \nabla \cdot (K\nabla p) = f(p),$$

using a mixed finite-element method. To linearize the mixed-method equations, we use a two grid scheme that relegates all the Newton-like iterations to a grid $\triangle_H$ much coarser than the original one $\triangle_h$, with no loss in order of accuracy so long as the mesh sizes obey $H = \mathcal{O}(\sqrt{h})$. The use of a multigrid-based solver for the indefinite linear systems that arise at each coarse-grid iteration, as well as for the similar system that arises on the fine grid, allows for even greater efficiency. © 1999 John Wiley & Sons, Inc. Numer Methods Partial Differential Eq 15: 317–332, 1999

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I. INTRODUCTION

Reaction-diffusion equations occur in numerous applications, noteworthy among them being mathematical models of hydrologic and bio-geochemical phenomena [1, 2]. The equations have the form

$$s\frac{\partial p}{\partial t} - \nabla \cdot (K\nabla p) = f(p), \qquad (\mathbf{x}, t) \in \Omega \times (0, T), \tag{1.1}$$

where Ω is a bounded domain in $\mathbb{R}^2$. They often arise from first-order mass balance and constitutive laws governing a *potential* p and a related *flux* $\mathbf{v}$:

$$\begin{aligned} s\frac{\partial p}{\partial t} + \nabla \cdot \mathbf{v} &= f(p) \qquad \text{mass balance;} \\ K^{-1}\mathbf{v} + \nabla p &= 0 \qquad \text{flux law.} \end{aligned} \tag{1.2}$$

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We assume that $p = 0$ on $\partial\Omega$, although our methods extend to nonzero Dirichlet and flux-type boundary conditions. In applications to groundwater hydrology, for example, the equations model water flow in an aquifer with leakage from a deformable confining rock layer lying above the aquifer. In this context, p is the unknown pressure head; s is a compressibility coefficient; K is the conductivity of the porous aquifer rock; $\mathbf{v}$ is the Darcy velocity of the water; and $f(p)$ models leakage.

Three numerical issues arise when we try to solve equations of this form. First, in some applications—such as those involving the transport of a contaminant in the fluid—it is important to approximate p and $\mathbf{v}$ with comparable accuracy. Second, to accommodate the nonlinear term $f(p)$, we must linearize via an efficient iterative scheme. Third, we must solve the resulting linear system effectively on fine spatial grids, especially when K is highly variable in space.

We examine an algorithm for numerically solving the system (1.2) that addresses all three issues. We use a mixed finite-element scheme to discretize the system in space with equal-order accuracy in p and $\mathbf{v}$. To linearize the resulting discrete equations, we use a two-grid scheme, which allows us to iterate on a grid much coarser than that used for the final solution. To solve the resulting linear systems, we use a multigrid-based "inner" iteration to reduce sensitivity to fine grids and variations in K.

Our approach follows several precedents. Mixed methods have attracted widespread attention among porous-media modelers for most of the past two decades [3, 4]. An important advantage of mixed methods is that they furnish a way to compute accurate approximations of both p and $\mathbf{v}$. Their main disadvantage is that they lead to large, indefinite linear systems. It is, therefore, of interest to find efficient solvers for the linear systems. Numerous investigators have attacked this problem [5–8]. We exploit a multigrid-based scheme first analyzed in [9].

We owe the impetus for using a two-grid approach to Xu [10], who demonstrates the possibility of mapping a fine-grid Galerkin finite-element problem onto a coarse grid. In that context, one solves the nonlinear problem via Newton-like iterations. After convergence on the coarse grid, one then extrapolates back to the fine grid using a Taylor expansion, as described below. Dawson and Wheeler [11] extend the method to mixed finite-element discretizations in which the coefficient K is nonlinear. The method discussed here treats the case when K is independent of p but f is not.

The remainder of the article is organized as follows: Section II describes the algorithm. Section III gives an error analysis establishing the method's convergence. Section IV presents some computational results confirming the method's utility, and Section V draws conclusions.

II. DESCRIPTION OF THE METHOD

A. Mixed Finite-Element Discretization

We begin by briefly reviewing the mixed finite-element discretization of the problem (1.2). On a rectangular domain $\Omega = (a, b) \times (c, d)$, let $\triangle^x = \{a = x_0 < x_1 < \cdots < x_m = b\}$ be a grid on the x-axis, and let $\triangle^y = \{c = y_0 < y_1 < \cdots < y_n = d\}$ be a grid on the y-axis. Thus, $\triangle^x \times \triangle^y$ is a rectangular grid on Ω with nodes$\{(x_i, y_j), 0 \le i \le m, 0 \le j \le n\}$. Let the mesh size of this rectangular grid be h. We denote this grid by $\triangle_h$ and call it the *fine grid*.

We use standard, lowest-order Raviart–Thomas [12] approximation subspaces W_h and Q_h for p and $\mathbf{v}$, respectively. The *potential space* W_h consists of functions that are piecewise constant on $\triangle_h$. The *flux space* Q_h has the form $Q_h^x \times Q_h^y$; here Q_h^x contains functions that are piecewise linear and continuous on $\triangle^x$ and piecewise constant on $\triangle^y$, while Q_h^y contains functions that are

piecewise constant on $\triangle^x$ and piecewise linear and continuous on $\triangle^y$. Figure 1 illustrates the grid structure and the locations of the nodal unknowns.

Given these spaces, we seek an approximate potential $p_h \in W_h$ and an approximate flux $\mathbf{v}_h \in Q_h$ such that

$$\begin{aligned} \int_\Omega K^{-1}\mathbf{v}_h \cdot \mathbf{q}_h - \int_\Omega p_h \nabla \cdot \mathbf{q}_h &= 0, && \text{for all} \quad \mathbf{q}_h \in Q_h, \\ \int_\Omega w_h \nabla \cdot \mathbf{v}_h + \int_\Omega w_h s \frac{\partial p_h}{\partial t} &= \int_\Omega w_h f(p_h), && \text{for all} \quad w_h \in W_h. \end{aligned} \tag{2.1}$$

Discretizing implicitly in time by the approximation $\partial p_h/\partial t \simeq (p_h^n - p_h^{n-1})/\Delta t$ converts the system (2.1) to a matrix equation for the vector $(V_h^n, P_h^n)^T$ of nodal degrees of freedom in $\mathbf{v}_h^n$ and p_h^n at each new time level n. Assuming a uniform time-step Δt, we obtain a block matrix equation

$$\begin{bmatrix} A_h & N_h \\ N_h^T & S_h \end{bmatrix} \begin{bmatrix} V_h^n \\ P_h^n \end{bmatrix} = \begin{bmatrix} 0 \\ F_h^n + S_h P_h^{n-1} \end{bmatrix}. \tag{2.2}$$

Here A_h is a symmetric, positive definite block containing integrals that involve K; N_h is essentially a differencing matrix; S_h is symmetric and positive definite and contains integrals involving s (and the factor $1/\Delta t$); and F_h^n contains the integrals $\int_\Omega w_h f(p_h^n)$.

Two remarks about this system are useful. First, although the block matrix on the left side of Eq. (2.2) is indefinite, its Schur complement $S_h - N_h^T A_h^{-1} N_h$ is symmetric and positive definite. Therefore, one can, in principle, eliminate the fluxes and solve the symmetric, positive definite, but nonsparse system

$$(S_h - N_h^T A_h^{-1} N_h) P_h^n = F_h^n + S_h P_h^{n-1}$$

for the potential. This fact helps motivate the multigrid-based scheme described below. Second, when K is scalar, one can use the trapezoid rule to compute the integrals $\int_\Omega K^{-1}\mathbf{v}_h \cdot \mathbf{q}_h$, yielding

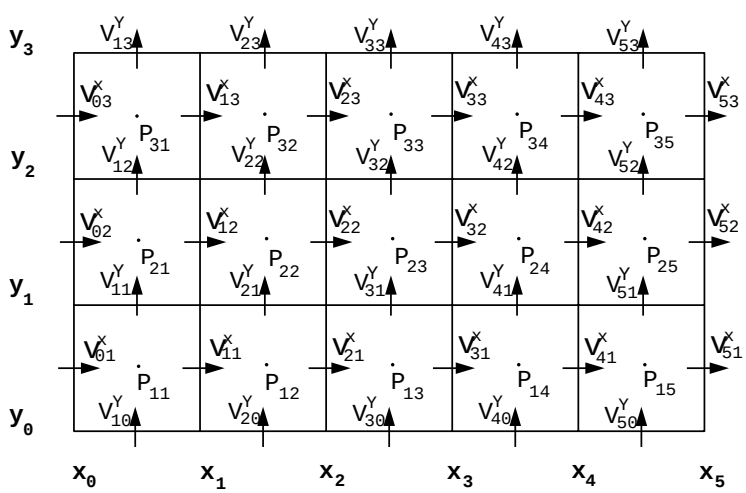

FIG. 1. A sample grid for the mixed finite element method.

a diagonal approximation to A_h but incurring no loss in order of accuracy [8]. For more about the matrix structure, see [9].

B. Linearization via the Two-Grid Method

To linearize the system (2.2), we use a two-grid method. The approach is analogous to those of Xu [10] and Dawson and Wheeler [11], cited earlier. The idea is to devote all of the effort of nonlinear iteration to coarse-grid problems. Thus, the iterations yield approximate finite-element solutions p_H^n, $\mathbf{v}_H^n$ on a coarse rectangular subgrid $\triangle_H \subset \triangle_h$ having mesh size $H > h$ and corresponding Raviart–Thomas spaces W_H, Q_H. These iterations follow a Newton-like heuristic,

$$\begin{aligned} p_H^{n,m} &= p_H^{n,m-1} + \delta p_H^{n,m}; \\ f(p_H^{n,m}) &\simeq f(p_H^{n,m-1}) + f'(p_H^{n,m-1})\delta p_H^{n,m}. \end{aligned} \tag{2.3}$$

The notation $(\cdot)^{n,m}$ signifies the mth iterate for the unknown at the nth time level. Here, $\delta p_H^{n,m}$ is an iterative increment to be determined by substituting into coarse-grid analogs of the system (2.1):

$$\begin{aligned} &\int_\Omega K^{-1}\mathbf{v}_H^{n,m}\cdot\mathbf{q}_H - \int_\Omega \delta p_H^{n,m}\nabla\cdot\mathbf{q}_H = \int_\Omega p_H^{n,m-1}\nabla\cdot\mathbf{q}_H, \quad \text{for all} \quad \mathbf{q}_H \in Q_H, \\ &\int_\Omega w_H \nabla\cdot v_H^{n,m} + \int_\Omega w_H\left[\frac{s}{\Delta t} - f'(p_H^{n,m-1})\right]\delta p_H^{n,m} \\ &\qquad = \int_\Omega w_H\left[f(p_H^{n,m-1}) - s\frac{p_H^{n,m-1} - p_H^{n-1}}{\Delta t}\right], \\ &\qquad\qquad \text{for all} \quad w_H \in W_H. \end{aligned} \tag{2.4}$$

To approximate p_h^n and $\mathbf{v}_h^n$ on $\triangle_h$, we then execute one iteration of the fine-grid system, using the converged coarse-grid solution p_H^n together with the heuristic

$$\begin{aligned} p_h^n \simeq \tilde{p}_h^n &= p_H^n + \delta p_h^n; \\ f(p_h^n) &\simeq f(p_H^n) + f'(p_H^n)\delta p_h^n. \end{aligned} \tag{2.5}$$

This tactic yields the following system of equations for the fine-grid solution:

$$\begin{aligned} &\int_\Omega K^{-1}\tilde{\mathbf{v}}_h^n\cdot\mathbf{q}_h - \int_\Omega \delta p_h^n \nabla\cdot\mathbf{q}_h = \int_\Omega p_H^n\nabla\cdot\mathbf{q}_h, \\ &\qquad\qquad \text{for all} \quad \mathbf{q}_h \in Q_h, \\ &\int_\Omega w_h\nabla\cdot\tilde{\mathbf{v}}_h^n + \int_\Omega w_h\left[s\frac{1}{\Delta t} - f'(p_H^n)\right]\delta p_h^n = \int_\Omega w_h\left[f(p_H^n) - s\frac{p_H^n - \tilde{p}_h^{n-1}}{\Delta t}\right], \\ &\qquad\qquad \text{for all} \quad w_h \in W_h. \end{aligned} \tag{2.6}$$

Thus, in block form, for the coarse-grid iterations, we solve linear systems of the form

$$\begin{bmatrix} A_H & N_H \\ N_H^T & S_H - \Psi_H^{n,m-1} \end{bmatrix} \begin{bmatrix} V_H^{n,m} \\ \delta P_H^{n,m} \end{bmatrix} = \begin{bmatrix} -N_H P_H^{n,m-1} \\ \overline{F}_H^{n,m-1} \end{bmatrix}, \tag{2.7}$$

where $\overline{F}_H^{n,m-1} = F_H^{n,m-1} - S_H(P_H^{n,m-1} - P_H^{n-1})$. Here, $\Psi_H^{n,m-1}$ is a matrix arising from integrals of the form

$$\int_\Omega w_H f'(p_H^{n,m-1}) \delta p_H^{n,m}.$$

For the single iteration on the fine grid, we solve a system of the form

$$\begin{bmatrix} A_h & N_h \\ N_h^T & S_h - \Psi_h^n \end{bmatrix} \begin{bmatrix} V_h^n \\ \delta P_h^n \end{bmatrix} = \begin{bmatrix} -N_h P_H^n \\ \overline{F}_h^n \end{bmatrix}, \tag{2.8}$$

where $\overline{F}_h^n = F_h^n - S_h(P_H^n - P_h^{n-1})$. Here, we compute P_H^n by using the interpolated coarse-grid solution on the fine grid, and the matrix Ψ_h^n arises from integrals of the form

$$\int_\Omega w_h f'(p_H^n) \delta p_h^n.$$

This linearization is efficient only if the coarse grid $\triangle_H$ can be significantly coarser than the fine grid $\triangle_h$ without sacrificing the accuracy associated with the fine-grid solution $(p_h^n, \mathbf{v}_h^n)$. The analysis in Section III demonstrates that, as with the Galerkin-based scheme proposed by Xu [10], one can choose $H = \mathcal{O}(\sqrt{h})$ and still retain the error estimates associated with the mixed finite-element scheme. Numerical examples in Section IV confirm that the errors in the fine-grid solutions are insensitive to the coarseness of $\triangle_H$ over a reasonable range of grid sizes.

C. Solving the Linear Systems

Whether one is iterating on the coarse grid or solving on the fine grid, one must solve sparse, symmetric, indefinite linear systems having the form

$$\begin{bmatrix} A & N \\ N^T & M \end{bmatrix} \begin{bmatrix} V \\ P \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}. \tag{2.9}$$

If D is a positive-definite diagonal approximation of A, then from the first block equation we have

$$DV + NP = F_1 + (D - A)V,$$

$$V = D^{-1}[F_1 + (D - A)V] - D^{-1}NP.$$

Substituting the last expression for V into the second block equation yields

$$\left(M - N^T D^{-1} N\right) P = F_2 - N^T D^{-1} F_1 - N^T D^{-1}(D - A)V. \tag{2.10}$$

Following earlier remarks in this section, the Schur complement $M - N^T D^{-1} N$ is symmetric and positive definite for small Δt. Moreover, there are reasonable choices for D: one can choose $D = \frac{2}{3}\,\mathrm{lump}\,(A)$, as in [9], or, when K is scalar, one can use the trapezoid rule to compute an approximation to A that is automatically diagonal [13]. In either case, $N^T D^{-1} N$ is sparse, having a pentadiagonal structure associated with standard five-point, cell-centered finite-difference approximations to Eq. (1.1).

Equation (2.10) suggests that we solve the linear system (2.9) using an inner iteration, whose levels we index by superscripts of the form (ℓ). To get from iterative level $\ell - 1$ to level ℓ, we calculate as follows:

(*i*) *Initialize the iterations:* $\ell \leftarrow 1$.
(*ii*) *Load the right-side vector:* $G^{(\ell-1)} \leftarrow F_2 - N^T D^{-1} F_1 - N^T D^{-1}(D - A)V^{(\ell-1)}$.
(*iii*) *Solve* $(M - N^T D^{-1} N)P^{(\ell)} = G^{(\ell-1)}$ *approximately for* $P^{(\ell)}$.
(*iv*) *Compute fluxes:* $V^{(\ell)} \leftarrow D^{-1}F_1 + D^{-1}(D - A)V^{(\ell-1)} - D^{-1}NP^{(\ell)}$.
(*v*) *If convergence test fails,* $\ell \leftarrow \ell + 1$.

Steps (*ii*) and (*iv*) require only sparse matrix multiplication. The inner step (*iii*) involves a standard, positive definite, pentadiagonal matrix equation, which we solve approximately using several V-cycles of a multigrid algorithm [14].

The following outline summarizes the flow of the overall solution procedure for a single time-step:

1. *Start new time-step* n.
2. *Begin nonlinear iterations on the coarse grid:* $m \leftarrow 1$.
    3. *Solve the linearized system (2.7) using the multigrid-based scheme:*
        4. *Execute (i) – (v) to get* $P_H^{n,m}$, $V_H^{n,m}$.
    5. *If nonlinear convergence fails,* $m \leftarrow m + 1$.
6. *Solve the fine-grid system (2.8): execute (i) – (v) for* P_h^n, V_h^n.
7. $n \leftarrow n + 1$.

III. CONVERGENCE ANALYSIS

Three theoretical results make this method effective. First, it is well known that the mixed-method discretization of the system (1.2) yields optimal-order error estimates for both potentials and fluxes [3]. Second, the multigrid-based linear solver is an efficient way to solve the matrix equations arising from two-grid linearization. In particular, the algorithm is relatively insensitive to the conditioning problems associated with fine grids and highly variable K. See [9] for more discussion of this result.

Third, the two-grid method affords a remarkably efficient linearization. It is possible to execute all the Newton-like iterations on very coarse grids, then use the heuristic (2.5) to obtain an accurate fine-grid solution in one additional step. We devote the rest of this section to analysis of the two-grid scheme. We begin by describing the discretization spaces and associated projection operators. We then review a convergence argument for the mixed method, using key parts of this argument in the analysis of the two-grid scheme.

A. Discretization and Projection Operators

In what follows, we assume that the reaction term f is twice continuously differentiable on Ω with bounded derivatives up to the second order and that K is a square integrable, symmetric, uniformly positive definite tensor defined on Ω. We denote the smallest and the largest eigenvalues of K by K_* and K^*, respectively. Thus, for any vector $\mathbf{y} \in \mathbb{R}^2$,

$$K_*|\mathbf{y}|^2 \leq \mathbf{y}^T K(\mathbf{x})\mathbf{y} \leq K^*|\mathbf{y}|^2, \qquad \text{for all} \quad \mathbf{x} \in \Omega. \tag{3.1}$$

The quantities of interest in the analysis are the potential p, $\mathbf{\Gamma} = -\nabla p$, and the flux $\mathbf{v} = K\mathbf{\Gamma}$. In this notation, $\nabla \cdot \mathbf{v} = -\nabla \cdot (K\nabla p)$. We use the following function spaces: $W = L^2(\Omega)$, with its usual inner product $(\cdot,\cdot)$ and norm $\|\cdot\|$, and

$$Q = H(\mathrm{div};\Omega) = \Big\{\mathbf{q} \in L^2(\Omega) \times L^2(\Omega) : \nabla \cdot \mathbf{q} \in L^2(\Omega)\Big\}.$$

The weak form for the first-order system (1.2) is as follows: find $p(\cdot,t) \in W$, $\mathbf{\Gamma}(\cdot,t) \in Q$, and $\mathbf{v}(\cdot,t) \in Q$ such that

$$\begin{aligned} (\partial p/\partial t,\ w) + (\nabla \cdot \mathbf{v},\ w) &= (f(p),\ w), && \text{for all} \quad w \in W, \\ (\mathbf{\Gamma}, \mathbf{q}) - (p, \nabla \cdot \mathbf{q}) &= 0, && \text{for all} \quad \mathbf{q} \in Q, \\ (\mathbf{v}, \mathbf{q}) - (K\mathbf{\Gamma}, \mathbf{q}) &= 0, && \text{for all} \quad \mathbf{q} \in Q. \end{aligned} \tag{3.2}$$

To analyze the mixed finite-element approximation to the system (3.2), let W_h and Q_h be finite-dimensional subspaces of W and Q, respectively, associated with the grid $\triangle_h$. Crucial for the analysis is the hypothesis that

$$\nabla \cdot \mathbf{q}_h \in W_h, \quad \text{for all} \quad \mathbf{q}_h \in Q_h. \tag{3.3}$$

This inclusion holds for the Raviart–Thomas spaces used here [12].

To analyze the temporal discretization on a time interval $(0,T)$, let $N > 0$, $\Delta t = T/N$, and $t^n = n\Delta t$. For functions $\varphi^n = \varphi(\cdot, t^n)$ that belong to $L^2(\Omega)$ at each time level n, we use the norm

$$\|\varphi\|_{l^2} := \Big(\sum_{n=1}^{N} \Delta t \|\varphi^n\|^2\Big)^{1/2}.$$

The analysis employs standard L^2 projections onto the spaces Q_h and W_h. Denote by $(\hat{p}_h^n, \hat{\mathbf{\Gamma}}_h^n) \in W_h \times Q_h$ the L^2 projection of $(p^n, \mathbf{\Gamma}^n)$, defined by the conditions

$$\begin{aligned} (p^n, w_h) &= (\hat{p}_h^n, w_h), && \text{for all} \quad w_h \in W_h, \\ (\mathbf{\Gamma}^n, \mathbf{q}_h) &= (\hat{\mathbf{\Gamma}}_h^n, \mathbf{q}_h), && \text{for all} \quad \mathbf{q}_h \in Q_h. \end{aligned} \tag{3.4}$$

We also need the projection $\Pi_h : H^1(\Omega) \times H^1(\Omega) \to Q_h$, defined by the condition

$$(\nabla \cdot (\Pi_h \mathbf{v}^n), \mathbf{q}_h) = (\nabla \cdot \mathbf{v}^n, \mathbf{q}_h), \qquad \text{for all} \quad \mathbf{q}_h \in Q_h. \tag{3.5}$$

These projections obey the following identities:

$$\begin{aligned} (\partial p^n/\partial t, w_h) &= (\partial \hat{p}_h^n/\partial t, w_h), && \text{for all} \quad w_h \in W_h, \\ (p^n, \nabla \cdot \mathbf{q}_h) &= (\hat{p}_h^n, \nabla \cdot \mathbf{q}_h), && \text{for all} \quad \mathbf{q}_h \in Q_h. \end{aligned} \tag{3.6}$$

The last identity holds by the condition (3.3).

The projections give rise to equations analogous to those in the weak form (3.2): approximating $\partial \hat{p}_h^n/\partial t$ using finite differences gives

$$\frac{1}{\Delta t}(\hat{p}_h^n - \hat{p}_h^{n-1}) = \frac{\partial \hat{p}_h^n}{\partial t} + \epsilon^n, \tag{3.7}$$

where ϵ^n is the truncation error. Substituting this approximation and the projection identities into Eqs. (3.2) yields the equations

$$\begin{aligned}
\frac{(\hat{p}_h^n - \hat{p}_h^{n-1},\ w_h)}{\Delta t} + (\nabla \cdot \Pi_h \mathbf{v}^n,\ w_h) &= (f(p^n),\ w_h) + (\epsilon^n, w_h), && \text{for all } w_h \in W_h, \\
(\hat{\mathbf{\Gamma}}_h^n, \mathbf{q}_h) - (\hat{p}_h^n, \nabla \cdot \mathbf{q}_h) &= 0, && \text{for all } \mathbf{q}_h \in Q_h, \\
(\Pi_h \mathbf{v}^n, \mathbf{q}_h) - (\Pi_h \mathbf{v}^n - \mathbf{v}^n, \mathbf{q}_h) + (K\mathbf{\Gamma}^n, \mathbf{q}_h) &= 0, && \text{for all } \mathbf{q}_h \in Q_h.
\end{aligned} \tag{3.8}$$

Corresponding to these equations are those governing the mixed finite-element approximation. We seek sequences $p_h^n \in W_h$, $\mathbf{\Gamma}_h^n \in Q_h$, and $\mathbf{v}_h^n \in Q_h$ such that

$$\begin{aligned}
\Delta t^{-1}(p_h^n - p_h^{n-1},\ w_h) + (\nabla \cdot \mathbf{v}_h^n,\ w_h) &= (f(p_h^n),\ w_h), && \text{for all } w_h \in W_h, \\
(\mathbf{\Gamma}_h^n, \mathbf{q}_h) - (p_h^n, \nabla \cdot \mathbf{q}_h) &= 0, && \text{for all } \mathbf{q}_h \in Q_h, \\
(\mathbf{v}_h^n, \mathbf{q}_h) - (K\mathbf{\Gamma}_h^n, \mathbf{q}_h) &= 0, && \text{for all } \mathbf{q}_h \in Q_h.
\end{aligned} \tag{3.9}$$

B. L^2 Error Estimates for the Mixed Method

We now estimate the errors $\|p^n - p_h^n\|$, $\|\mathbf{\Gamma}^n - \mathbf{\Gamma}_h^n\|$, and $\|\mathbf{v}^n - \mathbf{v}_h^n\|$. We assume that the finite-element spaces Q_h and W_h have the following approximation properties: For $w \in H^r(\Omega)$ and $\mathbf{q} \in H^r(\Omega) \times H^r(\Omega)$,

$$\begin{aligned}
\|\mathbf{q} - \Pi_h \mathbf{q}\| &\le C\|\mathbf{q}\|_r h^r, \qquad 1 \le r \le k+1, \\
\|w - \hat{w}_h\| &\le C\|w\|_r h^r, \qquad 1 \le r \le k+1,
\end{aligned} \tag{3.10}$$

where k is related to the polynomial degrees of $\mathbf{Q}_h$ and W_h. For the lowest-order Raviart–Thomas spaces used here, $k = 0$. Since $\|p^n - p_h^n\| \le \|p^n - \hat{p}_h^n\| + \|\hat{p}_h^n - p_h^n\|$, and similarly for $\mathbf{\Gamma}^n$ and $\mathbf{v}^n$, the approximation properties (3.9) reduce the problem to that of estimating $\|\hat{p}_h^n - p_h^n\|$, $\|\hat{\mathbf{\Gamma}}_h^n - \mathbf{\Gamma}_h^n\|$, and $\|\Pi_h \mathbf{v}^n - \mathbf{v}_h^n\|$. As $h \to 0$, the following theorem holds [11].

Theorem 3.1. *Let Q_h and W_h have the properties (3.3) and (3.9). Define $(p_h^n, \mathbf{\Gamma}_h^n, \mathbf{v}_h^n) \in W_h \times Q_h \times Q_h$ by (3.9) for $n \ge 1$. Then there exists a positive constant C, independent of h, such that*

$$\|p_h^N - p^N\| + \left[\sum_{n=1}^{N} \Delta t \|K^{1/2}(\mathbf{\Gamma}_h^n - \mathbf{\Gamma}^n)\|^2\right]^{1/2} \le C\left(h^{k+1} + \Delta t\right). \tag{3.11}$$

For the lowest-order spaces used in this article, $k = 0$, so the potential and flux errors are $\mathcal{O}(h + \Delta t)$.

The analysis follows the standard theory of mixed finite elements; we review it here, because we use parts of the argument in analyzing the two-grid scheme. Throughout, C denotes a positive constant, possibly different in different places but always independent of h and Δt.

Proof. Denote $\psi^n = \hat{p}_h^n - p_h^n$, $\gamma^n = \hat{\mathbf{\Gamma}}_h^n - \mathbf{\Gamma}_h^n$ and $\phi^n = \Pi_h \mathbf{v}^n - \mathbf{v}_h^n$. To derive error equations, we subtract Eqs. (3.9) from their respective counterparts (3.8). Then, in the results, we substitute the test functions $w_h = \psi^n$, $\mathbf{q}_h = \phi^n$, and $\mathbf{q}_h = \gamma^n$ in the first, second, and third

equations, respectively. We obtain

$$\Delta t^{-1}(\psi^n - \psi^{n-1},\ \psi^n) + (\nabla\cdot\phi^n,\ \psi^n) = (\epsilon^n,\ \psi^n) + (f(p^n) - f(p_h^n),\ \psi^n),$$

$$(\gamma^n, \phi^n) - (\psi^n, \nabla\cdot\phi^n) = 0, \tag{3.12}$$

$$(\phi^n, \gamma^n) = (\Pi_h \mathbf{v}^n - \mathbf{v}^n, \gamma^n) + (K(\mathbf{\Gamma}^n - \mathbf{\Gamma}_h^n), \gamma^n).$$

The goal is to reduce these equations to a single equation and then to bound the errors using a Gronwall-lemma argument.

At any point $\mathbf{x} \in \Omega$, by the Taylor theorem, $f(p^n(\mathbf{x})) - f(p_h^n(\mathbf{x})) = f'(\bar{p}^n(\mathbf{x}))(p^n(\mathbf{x}) - p_h^n(\mathbf{x}))$, for some value $\bar{p}^n(\mathbf{x})$. Therefore,

$$\begin{aligned}(f(p^n) - f(p_h^n),\ \psi^n) &= (f'(\bar{p}^n)(p^n - p_h^n),\ \psi^n)\\ &= (f'(\bar{p}^n)(p^n - \hat{p}_h^n),\ \psi^n) + (f'(\bar{p}^n)\psi^n,\ \psi^n).\end{aligned} \tag{3.13}$$

Also,

$$(K(\mathbf{\Gamma}^n - \mathbf{\Gamma}_h^n), \gamma^n) = (K(\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n), \gamma^n) + (K\gamma^n, \gamma^n). \tag{3.14}$$

Consequently, from Eq. (3.12),

$$\begin{aligned}\Delta t^{-1}(\psi^n - \psi^{n-1},\ \psi^n) &+ (K\gamma^n, \gamma^n)\\ &= (\epsilon^n,\ \psi^n) + (f'(\bar{p}^n)(p^n - \hat{p}_h^n),\ \psi^n)\\ &\quad +(f'(\bar{p}^n)\psi^n,\ \psi^n) - (\Pi_h \mathbf{v}^n - \mathbf{v}^n, \gamma^n)\\ &\quad -(K(\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n), \gamma^n).\end{aligned} \tag{3.15}$$

We now use the Gronwall lemma to estimate the left side. Denote $F = \|f'(p)\|_\infty$. By the inequality $-(\|\psi^{n-1}\|^2 + \|\psi^n\|^2)/2 \le -(\psi^{n-1}, \psi^n)$ and the identity $(K\gamma^n,\ \gamma^n) = \|K^{1/2}\gamma^n\|^2$, the following inequality holds for any positive number δ:

$$\begin{aligned}\frac{1}{2\Delta t}(\|\psi^n\|^2 - \|\psi^{n-1}\|^2) &+ \|K^{1/2}\gamma^n\|^2\\ &\le \frac{1}{2\delta}\|\Pi_h \mathbf{v}^n - \mathbf{v}^n\|^2 + \frac{\delta}{2}\|\gamma^n\|^2 + \frac{1}{2\delta}K^*\|\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n\|^2 + \frac{\delta}{2}\|\gamma^n\|^2\\ &\quad +\frac{1}{2}\|\epsilon^n\|^2 + \frac{1}{2}\|\psi^n\|^2 + \frac{1}{2}F\|p^n - \hat{p}_h^n\|^2 + \frac{1}{2}\|\psi^n\|^2 + F\|\psi^n\|^2\\ &= \frac{1}{2\delta}\|\Pi_h \mathbf{v}^n - \mathbf{v}^n\|^2 + \delta\|\gamma^n\|^2 + \frac{1}{2\delta}K^*\|\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n\|^2\\ &\quad +\frac{1}{2}\|\epsilon^n\|^2 + \frac{1}{2}F\|p^n - \hat{p}_h^n\|^2 + (1+F)\|\psi^n\|^2.\end{aligned} \tag{3.16}$$

When $\delta \le K_*/2$, we have $\delta\|\gamma^n\|^2 \le \frac{1}{2}\|K^{1/2}\gamma^n\|^2$. In this case,

$$\begin{aligned}\frac{1}{2\Delta t}(\|\psi^n\|^2 - \|\psi^{n-1}\|^2) \;+\; & \frac{1}{2}\|K^{1/2}\gamma^n\|^2 \\ \le\; & \frac{1}{2\delta}\|\Pi_h\mathbf{v}^n - \mathbf{v}^n\|^2 + \frac{1}{2\delta}K^*\|\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n\|^2 \\ & + \frac{1}{2}\|\epsilon^n\|^2 + \frac{1}{2}F\|p^n - \hat{p}_h^n\|^2 + (1+F)\|\psi^n\|^2.\end{aligned} \tag{3.17}$$

Multiplying both sides of (3.17) by $2\Delta t$ and summing over n, $n = 1, 2, \ldots, N$, we get

$$\begin{aligned}(\|\psi^N\|^2 - \|\psi^0\|^2) \;+\; & \sum_{n=1}^{N}\Delta t\|K^{1/2}\gamma^n\|^2 \\ \le\; & \frac{1}{\delta}\sum_{n=1}^{N}\Delta t\|\Pi_h\mathbf{v}^n - \mathbf{v}^n\|^2 + \frac{K^*}{\delta}\sum_{n=1}^{N}\Delta t\|\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n\|^2 + \Delta t\sum_{n=1}^{N}\|\epsilon^n\|^2 \\ & + F\sum_{n=1}^{N}\Delta t\|p^n - \hat{p}_h^n\|^2 + 2(1+F)\sum_{n=1}^{N}\Delta t\|\psi^n\|^2.\end{aligned} \tag{3.18}$$

But, with proper choice of the initial function p_h^0, we have $\psi^0 = 0$, and it follows that

$$\begin{aligned}\|\psi^N\|^2 \;+\; & \sum_{n=1}^{N}\Delta t\|K^{1/2}\gamma^n\|^2 \\ \le\; & \Delta t\sum_{n=1}^{N}C\Big(\|\Pi_h\mathbf{v}^n - \mathbf{v}^n\|^2 + \|\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n\|^2 + \|p^n - \hat{p}_h^n\|^2\Big) \\ & + 2\Delta t(1+F)\sum_{n=1}^{N}\|\psi^n\|^2 + \Delta t\sum_{n=1}^{N}\|\epsilon^n\|^2.\end{aligned} \tag{3.19}$$

When $\Delta t \le 1/[2(1+F)]$, the Gronwall lemma yields

$$\begin{aligned}\|\psi^N\|^2 \;+\; & \sum_{n=1}^{N}\Delta t\|K^{1/2}\gamma^n\|^2 \\ \le\; & C\sum_{n=1}^{N}\Delta t\Big(\|\underbrace{\Pi_h\mathbf{v}^n - \mathbf{v}^n}_{\le Ch^{k+1}}\|^2 + \|\underbrace{\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n}_{\le Ch^{k+1}}\|^2 + \|\underbrace{p^n - \hat{p}_h^n}_{\le Ch^{k+1}}\|^2\Big) + C\Delta t^2 \\ \le\; & C(h^{k+1})^2 + C\Delta t^2.\end{aligned} \tag{3.20}$$

Therefore,

$$\|\psi^N\| + \|K^{1/2}\gamma\|_{l^2} \le C\left(h^{k+1} + \Delta t\right). \tag{3.21}$$

By the approximation properties (3.9) and the triangle inequalities $\|p_h^N - p^N\| \le \|\psi^N\| + \|\hat{p}_h^N - p^N\|$ and $\|K^{1/2}(\mathbf{\Gamma}_h^n - \mathbf{\Gamma}^n)\| \le \|K^{1/2}\gamma^n\| + \|K^{1/2}(\hat{\mathbf{\Gamma}}_h^n - \mathbf{\Gamma}^n)\|$, we conclude that

$$\begin{aligned}
\|p_h^N - p^N\| + \|K^{1/2}(\mathbf{\Gamma}_h^n - \quad \mathbf{\Gamma}^n)\|_{l^2} & \\
\le \quad & \underbrace{\|\psi^N\| + \|K^{1/2}\gamma\|_{l^2}}_{\substack{\le C(h^{k+1}+\Delta t) \\ \text{by Gronwall} \\ \text{inequality}}} \quad + \quad \underbrace{\|\hat{p}_h^N - p^N\|}_{\substack{\le Ch^{k+1} \\ \text{by approximation} \\ \text{properties of } W_h}} \\
& + \quad \underbrace{\|K^{1/2}(\hat{\mathbf{\Gamma}}_h^n - \mathbf{\Gamma}^n)\|_{l^2}.}_{\substack{\le Ch^{k+1} \\ \text{by approximation properties} \\ \text{of } Q_h \text{ and bounds on } K}}
\end{aligned} \tag{3.22}$$

This inequality establishes the familiar convergence result for the mixed finite-element method. ∎

C. Error Estimates for the Two-Grid Method

We now analyze the two-grid method for iteratively solving the nonlinear system (3.9). The method has two stages:

Stage 1: On the coarse grid $\triangle_H$, solve the following nonlinear system for $(p_H^n,\ \mathbf{\Gamma}_H^n,\ \mathbf{v}_H^n) \in (W_H, Q_H, Q_H)$:

$$\begin{aligned}
\Delta t^{-1}(p_H^n - p_H^{n-1},\ w_H) + (\nabla\cdot\mathbf{v}_H^n,\ w_H) &= (f(p_H^n),\ w_H), && \text{for all} \quad w_H \in W_H, \\
(\mathbf{\Gamma}_H^n, \mathbf{q}_H) - (p_H^n, \nabla\cdot\mathbf{q}_H) &= 0, && \text{for all} \quad \mathbf{q}_H \in Q_H, \\
(\mathbf{v}_H^n, \mathbf{q}_H) - (K\mathbf{\Gamma}_H^n, \mathbf{q}_H) &= 0, && \text{for all} \quad \mathbf{q}_H \in Q_H.
\end{aligned} \tag{3.23}$$

Stage 2: Then, on the fine grid $\triangle_h$, solve the following linear system for $(\tilde{p}_h^n, \tilde{\mathbf{\Gamma}}_h^n, \tilde{\mathbf{v}}_h^n) \in W_h \times Q_h \times Q_h$:

$$\begin{aligned}
\frac{(\tilde{p}_h^n - \tilde{p}_h^{n-1},\ w_h)}{\Delta t} + (\nabla\cdot\tilde{\mathbf{v}}_h^n,\ w_h) &= (f(p_H^n) + f'(p_H^n)(\tilde{p}_h^n - p_H^n),\ w_h), && \text{for all } w_h \in W_h \\
(\tilde{\mathbf{\Gamma}}_h^n, \mathbf{q}_h) - (\tilde{p}_h^n, \nabla\cdot\mathbf{q}_h) &= 0, && \text{for all } \mathbf{q}_h \in Q_h, \\
(\tilde{\mathbf{v}}_h^n, \mathbf{q}_h) - (K\tilde{\mathbf{\Gamma}}_h^n, \mathbf{q}_h) &= 0, && \text{for all } \mathbf{q}_h \in Q_h.
\end{aligned} \tag{3.24}$$

This scheme is attractive, because it requires nonlinear iteration only on the coarse grid, giving the fine-grid solution in a single, linear, Newton-like step. However, it incurs additional errors of the forms $\|p_h^n - \tilde{p}_h^n\|$, $\|\mathbf{\Gamma}_h^n - \tilde{\mathbf{\Gamma}}_h^n\|$, and $\|\mathbf{v}_h^n - \tilde{\mathbf{v}}_h^n\|$ with respect to the true finite-element solution discussed above. The following theorem estimates these errors.

Theorem 3.2. *Let Q_h and W_h have the properties (3.3) and (3.9). Let $(\tilde{p}_h^n, \tilde{\mathbf{\Gamma}}_h^n, \tilde{\mathbf{v}}_h^n) \in W_h \times Q_h \times Q_h$ satisfy (3.8) for $n \geq 1$. Then there exists a positive constant C, independent of h, such that*

$$\|\tilde{p}_h^N - p^N\| + \left[\sum_{n=1}^{N} \Delta t \|K^{1/2}(\tilde{\mathbf{\Gamma}}_h^n - \mathbf{\Gamma}^n)\|^2 \right]^{1/2} \leq C\left(h^{k+1} + H^{2k+2} + \Delta t\right). \quad (3.25)$$

Proof. To estimate these errors, set $\xi^n = \hat{p}_h^n - \tilde{p}_h^n$, $\upsilon^n = \hat{\mathbf{\Gamma}}_h^n - \tilde{\mathbf{\Gamma}}_h^n$, and $\theta^n = \Pi_h \mathbf{v}^n - \tilde{\mathbf{v}}_h^n$. Subtracting Eqs. (3.24) from their respective counterparts (3.8) and choosing $w_h = \xi^n$, $\mathbf{q}_h = \theta^n$, and $\mathbf{q}_h = \upsilon^n$, we have

$$\begin{aligned}
\frac{1}{\Delta t}(\xi^n - \xi^{n-1},\ \xi^n) + (\nabla \cdot \theta^n,\ \xi^n) &= (\epsilon^n,\ \xi^n) + (f(p^n),\ \xi^n) \\
&\quad -(f(p_H^n) + f'(p_H^n)(\tilde{p}_h^n - p_H^n), \xi^n), \\
(\upsilon^n, \theta^n) - (\xi^n, \nabla \cdot \theta^n) &= 0, \\
(\theta^n, \upsilon^n) &= (\Pi_h \mathbf{v}^n - \mathbf{v}^n, \upsilon^n) + (K(\mathbf{\Gamma}^n - \tilde{\mathbf{\Gamma}}_h^n), \upsilon^n).
\end{aligned} \quad (3.26)$$

We now rewrite terms on the right side. In the first equation, a Taylor expansion of f about p_H^n yields

$$\begin{aligned}
(f(p^n), \xi^n) &- (f(p_H^n) + f'(p_H^n)(\tilde{p}_h^n - p_H^n), \xi^n) \\
&= (f(p_H^n) + f'(p_H^n)(p^n - p_H^n) + \tfrac{1}{2} f''(\bar{p})(p^n - p_H^n)^2, \xi^n) \\
&\quad -(f(p_H^n) + f'(p_H^n)(\tilde{p}_h^n - p_H^n), \xi^n) \\
&= (f'(p_H^n)(p^n - \tilde{p}_h^n), \xi^n) + (\tfrac{1}{2} f''(\bar{p})(p^n - p_H^n)^2, \xi^n) \\
&= (f'(p_H^n)(p^n - \hat{p}_h^n), \xi^n) + (f'(p_H^n)\xi^n, \xi^n) + (\tfrac{1}{2} f''(\bar{p})(p^n - p_H^n)^2, \xi^n),
\end{aligned} \quad (3.27)$$

for some function $\bar{p}$. In the third equation,

$$(K(\mathbf{\Gamma}^n - \tilde{\mathbf{\Gamma}}_h^n), \upsilon^n) = (K(\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n), \upsilon^n) + (K\upsilon^n, \upsilon^n). \quad (3.28)$$

Substituting these expressions and eliminating θ^n yield

$$\begin{aligned}
\frac{1}{\Delta t}(\xi^n - \xi^{n-1}, \xi^n) &+ (K\upsilon^n, \upsilon^n) \\
&= (\epsilon^n, \xi^n) + (f'(p_H^n)(p^n - \hat{p}_h^n), \xi^n) + (f'(p_H^n)\xi^n, \xi^n) \\
&\quad +(\frac{1}{2} f''(\bar{p})(p^n - p_H^n)^2, \xi^n) - (\Pi_h \mathbf{v}^n - \mathbf{v}^n, \upsilon^n) - (K(\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n), \upsilon^n).
\end{aligned} \quad (3.29)$$

If we denote $G = \|f''\|_\infty$, then arguments similar to those used for the finite-element error estimates yield

$$\begin{aligned}
\frac{1}{2\Delta t}(\|\xi^n\|^2 - \|\xi^{n-1}\|^2) \;+\; & \|K^{1/2}\upsilon^n\|^2 \\
\leq\; & \frac{1}{2}\|\epsilon^n\|^2 + \frac{1}{2}\|\xi^n\|^2 + \frac{1}{2}F\|p^n - \hat{p}_h^n\|^2 + \frac{1}{2}\|\xi^n\|^2 \\
& + F\|\xi^n\|^2 + G\|(p^n - p_H^n)^2\|^2 + \frac{1}{2}\|\xi^n\|^2 \\
& + \frac{1}{2\delta}\|\Pi_h\mathbf{v}^n - \mathbf{v}^n\|^2 + \frac{\delta}{2}\|\upsilon^n\|^2 + \frac{K^*}{2\delta}\|\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n\|^2 + \frac{\delta}{2}\|\upsilon^n\|^2 \\
=\; & \frac{1}{2}\|\epsilon^n\|^2 + \frac{3}{2}\|\xi^n\|^2 + \delta\|\upsilon^n\|^2 + G\|(p^n - p_H^n)^2\|^2 \\
& + C\Big(\|\Pi_h\mathbf{v}^n - \mathbf{v}^n\|^2 + \|\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n\|^2 + \|p^n - \hat{p}_h^n\|^2\Big),
\end{aligned} \tag{3.30}$$

for any positive constant δ. Choosing $\delta \leq K_*/2$ yields

$$\begin{aligned}
\frac{1}{2\Delta t}(\|\xi^n\|^2 - \|\xi^{n-1}\|^2) \;+\; & \frac{1}{2}\|K^{1/2}\upsilon^n\|^2 \\
\leq\; & \frac{1}{2}\|\epsilon^n\|^2 + \frac{3}{2}\|\xi^n\|^2 + G\|(p^n - p_H^n)^2\|^2 \\
& + C\Big(\|\Pi_h\mathbf{v}^n - \mathbf{v}^n\|^2 + \|\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n\|^2 + \|p^n - \hat{p}_h^n\|^2\Big).
\end{aligned} \tag{3.31}$$

Now multiplying both sides of (3.31) by $2\Delta t$ and summing over n for $n = 1, 2, \ldots, M$ yield

$$\begin{aligned}
(\|\xi^M\|^2 - \|\xi^0\|^2) \;+\; & \sum_{n=1}^{M}\Delta t\|K^{1/2}\upsilon^n\|^2 \\
\leq\; & \sum_{n=1}^{M}\Delta t\|\epsilon^n\|^2 + \sum_{n=1}^{M}3\Delta t\|\xi^n\|^2 \\
& + C\Bigg[\sum_{n=1}^{M}\Delta t\|(p^n - p_H^n)^2\|^2 + \sum_{n=1}^{M}\Delta t\|p^n - \hat{p}_h^n\|^2 \\
& + \sum_{n=1}^{M}\Delta t\|\Pi_h\mathbf{v}^n - \mathbf{v}^n\|^2 + \sum_{n=1}^{M}\Delta t\|\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n\|^2\Bigg],
\end{aligned} \tag{3.32}$$

for some positive number C, independent of mesh size h. Since $\xi^0 = 0$, the Gronwall lemma gives

$$\|\xi^M\|^2 + \sum_{n=1}^{M} \Delta t \|K^{1/2} \upsilon^n\|^2 \leq C \sum_{n=1}^{M} \Delta t \Big[\|(p^n - p_H^n)^2\|^2 + \|\Pi_h \mathbf{v}^n - \mathbf{v}^n\|^2 + \|\mathbf{\Gamma}^n - \hat{\mathbf{\Gamma}}_h^n\|^2 + \|p^n - \hat{p}_h^n\|^2 \Big] + C\Delta t^2. \tag{3.33}$$

Thus, from Theorem 3.1 and approximation properties (3.9), we complete the proof of the main convergence theorem. ∎

Theorem 3.2 demonstrates a remarkable fact about the two-grid scheme: the approximation space $W_H \times \mathbf{Q}_H \times \mathbf{Q}_H$ can be very coarse ($H >> h$), and we can still maintain accuracy. Specifically, for the Raviart–Thomas spaces used here, $k = 0$, and the L^2 errors in the fine-grid pressure and velocity are $\mathcal{O}(h + H^2 + \Delta t)$. Hence, in two space dimensions, the nonlinear iterations can involve as few as $\mathcal{O}(H^{-2}) = \mathcal{O}(h^{-1})$ unknowns, even though the original nonlinear system on the fine grid involves $\mathcal{O}(h^{-2})$ unknowns. We confirm this result numerically in the next section.

IV. NUMERICAL EXAMPLES

To illustrate the effectiveness of the two-grid linearization, we examine the following simple test problem:

$$\frac{\partial p}{\partial t} - \nabla \cdot (K \nabla p) = -2p - \frac{1}{2}p^2 + g(x, y), \qquad \text{for} \quad t > 0,\ \Omega = (0,1)^2, \tag{4.1}$$

with $p(x, y, 0) = 0$ and $p = 0$ on $\partial\Omega$. Here, $K(x, y) = x^2 + y^2 + 1$, the function $g(x, y)$ being chosen so that the exact solution is $p(x, y, t) = -t \exp(x + y)x(x - 1)y(y - 1)$. We use $\Delta t = 0.1$ throughout. Table I shows how the L^2 errors in $\tilde{p}_h$ and $\tilde{\mathbf{v}}_h$ at $t = 0.5$ shrink with h when $H = h^{1/2}$. Both the pressure and velocity errors decrease linearly as $h \to 0$, even though the nonlinear iterations occur on a grid much coarser than h.

Theorem 3.2 implies that we can iterate on a very coarse grid and still get good approximations by taking one iteration on the fine grid. For the same test problem as above, we fix the mesh size of the fine grid as $h = 2^{-7}$, then vary the mesh size H of the coarse-grid. Table II shows how the L^2 errors in $\tilde{p}_h$ and $\tilde{\mathbf{v}}_h$ vary with H at $t = 0.5$. Over the range of coarse grids tested, the

TABLE I. Decay in L^2-errors with grid refinement when $H = h^{1/2}$.

h	H	$\frac{\|\tilde{p}_h - p\|}{\|p\|}$	$\frac{\|\tilde{\mathbf{v}}_h - \mathbf{v}\|}{\|\mathbf{v}\|}$
2^{-2}	2^{-1}	4.401×10^{-2}	7.834×10^{-2}
2^{-4}	2^{-2}	3.491×10^{-3}	5.403×10^{-3}
2^{-6}	2^{-3}	2.467×10^{-4}	3.348×10^{-4}
2^{-8}	2^{-4}	1.602×10^{-5}	2.603×10^{-5}

TABLE II. L^2 errors for $h = 2^{-7}$ vs. coarse mesh size H.

H	$\frac{\|\tilde{p}_h - p\|}{\|p\|}$	$\frac{\|\tilde{\mathbf{v}}_h - \mathbf{v}\|}{\|\mathbf{v}\|}$	CPU seconds
2^{-1}	9.533×10^{-5}	1.809×10^{-4}	168.20
2^{-2}	6.416×10^{-5}	1.110×10^{-4}	165.19
2^{-3}	6.341×10^{-5}	8.970×10^{-5}	164.05
2^{-4}	6.319×10^{-5}	8.887×10^{-5}	166.44
2^{-5}	6.310×10^{-5}	8.887×10^{-5}	194.84
2^{-6}	6.307×10^{-5}	9.009×10^{-5}	344.44
2^{-7}	6.303×10^{-5}	8.198×10^{-5}	608.42

fine-grid errors are quite insensitive to the coarseness of the grid on which we iterate. The table also shows estimates of total CPU time on an SGI Indy workstation, although we made no special attempt to optimize the code.

Although the CPU times reported here provide only a rough measure, they give some indication of the efficiency of the two-grid approach. There is a significant range of coarse-grid mesh sizes over which the runtime efficiency and accuracy of the method is fairly insensitive. But choosing too fine a coarse grid sacrifices efficiency without improving accuracy. By far the longest runtime occurs when we use no coarse-grid iterations, that is, when $H = h$.

V. CONCLUSIONS

Previous work [9, 14] illustrates the utility of the multigrid-based linear solution scheme for the mixed finite-element system (2.9). The two-grid linearization offers an attractive way to extend this approach to nonlinear problems involving reaction-diffusion equations. The key feature of the two-grid method is that it allows one to execute all the nonlinear iterations on a system associated with a coarse spatial grid, without sacrificing the order of accuracy of the fine-grid solution.

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